# Abhinav Painuli

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## Education

**Cluster Innovation Centre (University of Delhi)** Delhi, India
*Bachelors in Technology in Information Technology and Mathematical Innovation* *2022 – 2026*
*Current GPA: 8.44*

## Experience

**NiyamR AI** Dec 2025 – Present
*Full-Stack AI Intern* *Remote*

- Developed a document intelligence web application to process and analyze 1000+ financial documents.
- Designed a scalable end-to-end AI pipeline for document ingestion, indexing, and retrieval, deployed on AWS for reliability and efficiency.
- Integrated hybrid RAG-based verification and semantic search to enhance transaction matching accuracy and compliance validation.
- Developed Third Prism, a multi-agent AI prediction engine leveraging OASIS architecture and Knowledge Graphs, with autonomous agent swarms for dynamic goals and interactions.

**IIT Bombay** Oct 2024 – May 2025
*Machine Learning Intern* *Remote*

- Fine-tuned multiple large language models (LLMs) from Hugging Face, adapting models for performance optimization on custom datasets.
- Developed Automatic Speech Recognition pipelines for multilingual audio transcription using Speech2Text models, integrating diarization, Voice Activity Detection (VAD) and XML Generation.
- Developed ASR Transcripts error detection pipelines and built pipeline integrating diarization and transcription.
- Developed **SpeechQMAgent**, a multi-agent framework for automated verification of large-scale multilingual speech-text datasets with **22** quality checks.

## Research and Achievements

- Second author on **Rolling Out Data Quality Overnight, without losing the plot: A Multi-Agent System for Speech Data Quality Management** accepted to **Findings of ACL 2026**
- Acknowledged in EACL 2026 paper: **Post-ASR Correction in Hindi: Comparing Language Models and Large Language Models in Low-Resource Scenarios**.
- Finalist, **Hack the Future 2025 (IIT Gandhinagar)** – selected among top teams nationwide for AI solution.

## Projects

**AarogyaAI: AI-Powered Telemedicine Platform** | *Python, React, Cartesia* GitHub

- Built a multi-stage STT → LLM reasoning pipeline using Cerebras AI and Llama 3.1 8B, optimized for real-time inference and low-latency clinical note generation.
- Integrated Cartesia STT and tuned prompt structures + context windows to reduce decoding overhead and speed up structured medical output generation.
- Engineered multi-agent orchestration with context retention, safety filtering, and medical entity extraction for efficient, reliable LLM-based reasoning.
- Designed Next.js dashboards and assistants; containerized the entire inference stack using Docker for reproducible deployments.

**Semantic Knowledge-Augmented RAG for Improved Question-Answering** | *Python, ChromaDB* GitHub

- Built a semantic knowledge-augmented RAG system with hybrid retrieval using vector search and BM25.
- Implemented semantic chunking and lightweight knowledge graph construction using NER and dependency parsing.
- Enhanced answer grounding through graph-aware context expansion and constrained LLM prompting.